

5CM506 Data driven systems – Tutorial

Microsoft Access I

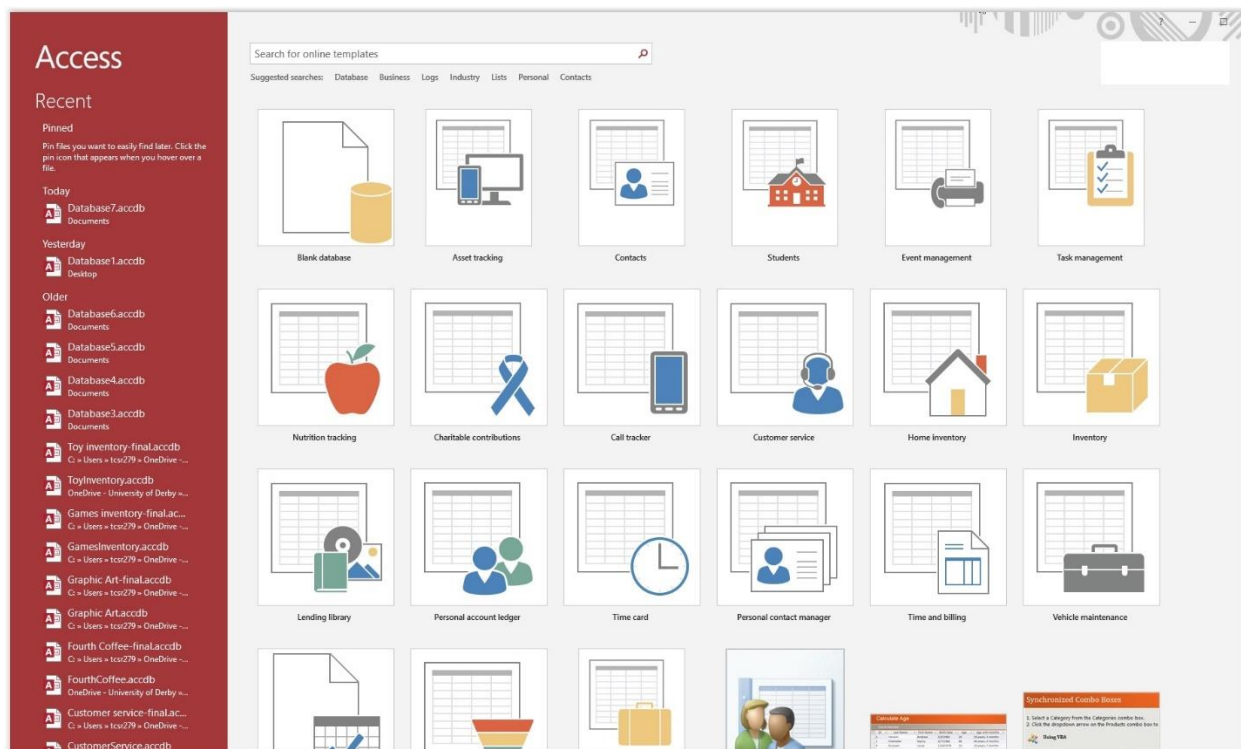
Creating a Database

This tutorial series is designed to introduce you to relational database fundamentals using Microsoft Access, a popular desktop DBMS.

This week's session will cover using Microsoft Access to implement a simple database. Obviously, Microsoft Access isn't an appropriate database management system for building industrial-strength systems, but it does all the same things they do – though Access lacks industrial-strength multi-user capability – so it's a great way to explore database fundamentals.

To start Access on the lab machines, select “Start | Access 2019”¹.

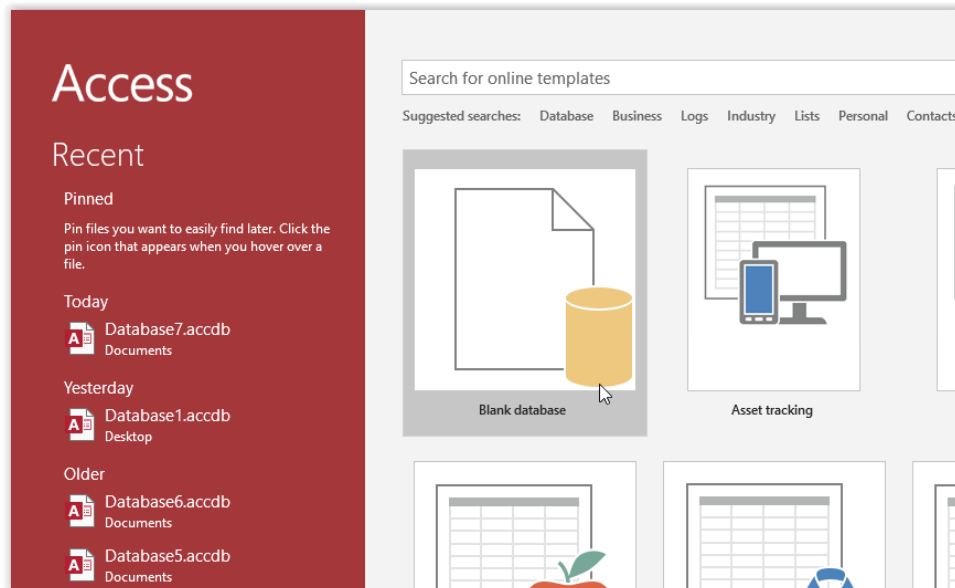
You should see something (roughly... depending on Access version) like the following:



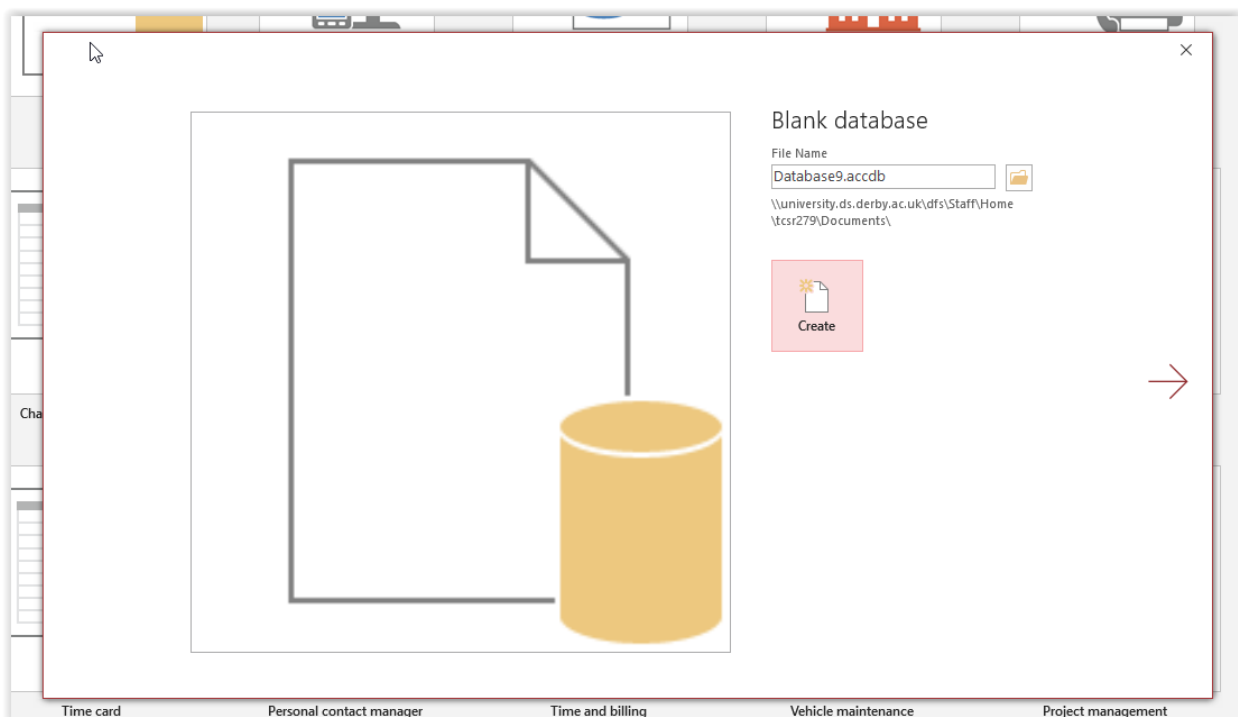
¹ When this document was originally written, it was written in an earlier version of Office. We've been through Office 2013, 2016 and now Office 2019 is the current version. Depending on where you are in the university, you might see different versions, and some of you will have a different version at home. Indeed, the screen shots taken in this document use Access for Office 365. Whilst the appearance may be slightly different between versions, the functionality is the same. If Microsoft Access isn't where I said it was, look through the Start menus until you find it. Any version is fine. If you can't find it, ask your tutor or see if one of your fellow students can help.

Let's imagine we're going to create a database to record data about something you should all be familiar with: student records in a university. In particular, we're going to record information about students, courses, and modules.

We'll want to create a new database. To do that, click on the "Blank Database" icon, as shown below:



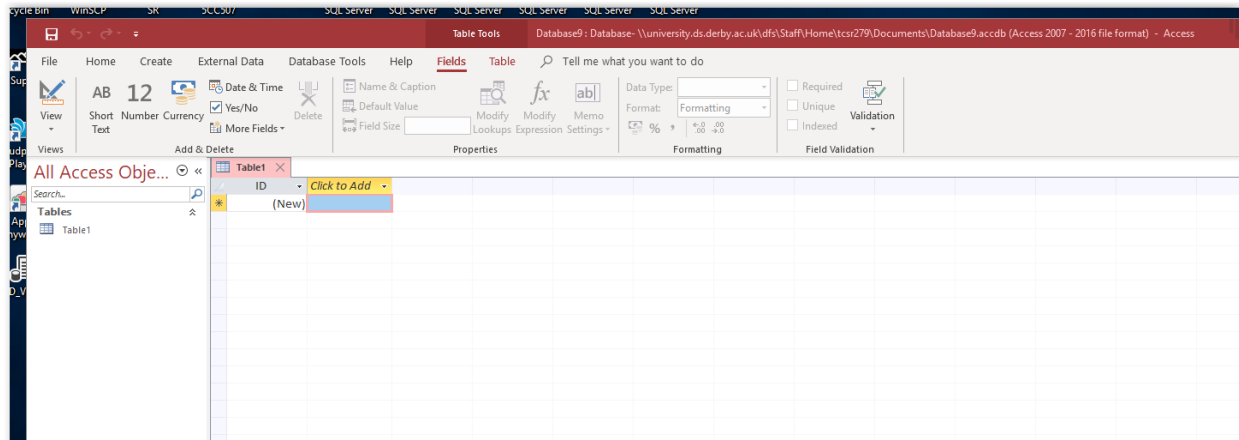
When you click on the button, Access will either ask you for the name of a new database, as shown below:



Or, let Access choose a default name for you. You may change it to whatever name you like, but it must obey the usual Windows file name conventions and it must have the .accdb extension.

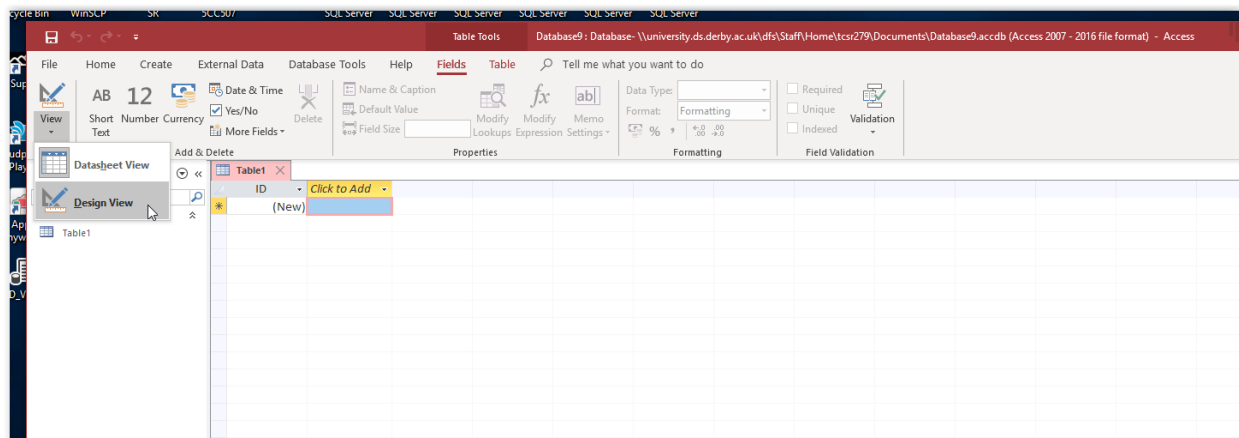
If necessary, press the Create button once you've chosen an appropriate name. Depending on the version of Access you're using, it might jump directly to the content shown below:

Access will now start to create a table for you in “Datasheet” mode and is asking you to enter data. It will look like the following:

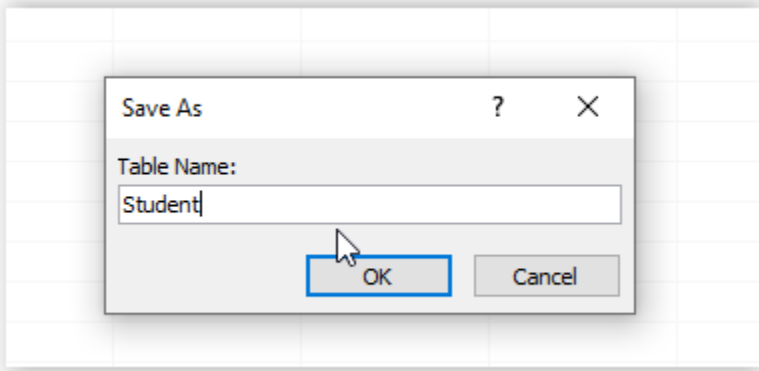


However, to properly complete the table definition, we need to put Access into the “Design View”.

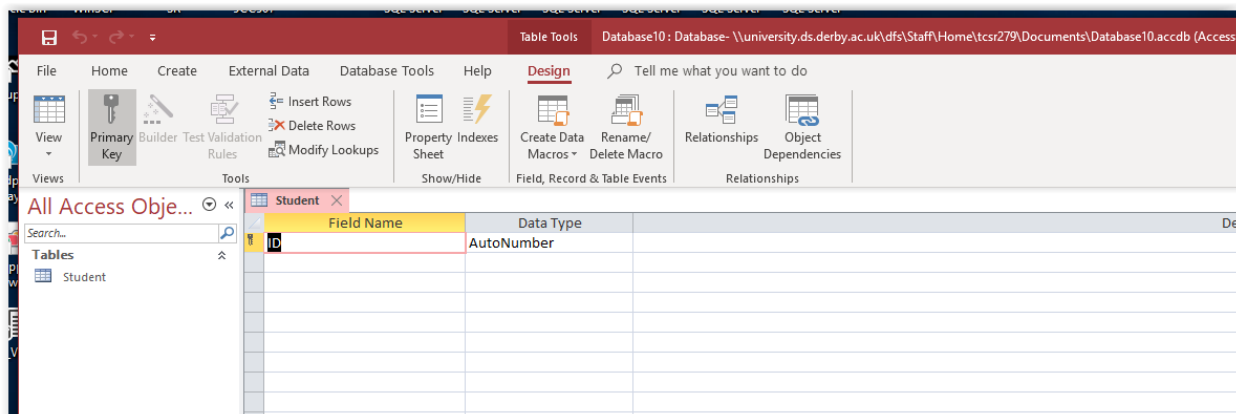
Click on the View icon toward the upper left-hand corner of the window, and click on “Design View”, as shown below:



If Access asks you to save the table definition before continuing, give it the name “Students” and save it.



Now you should see the following:



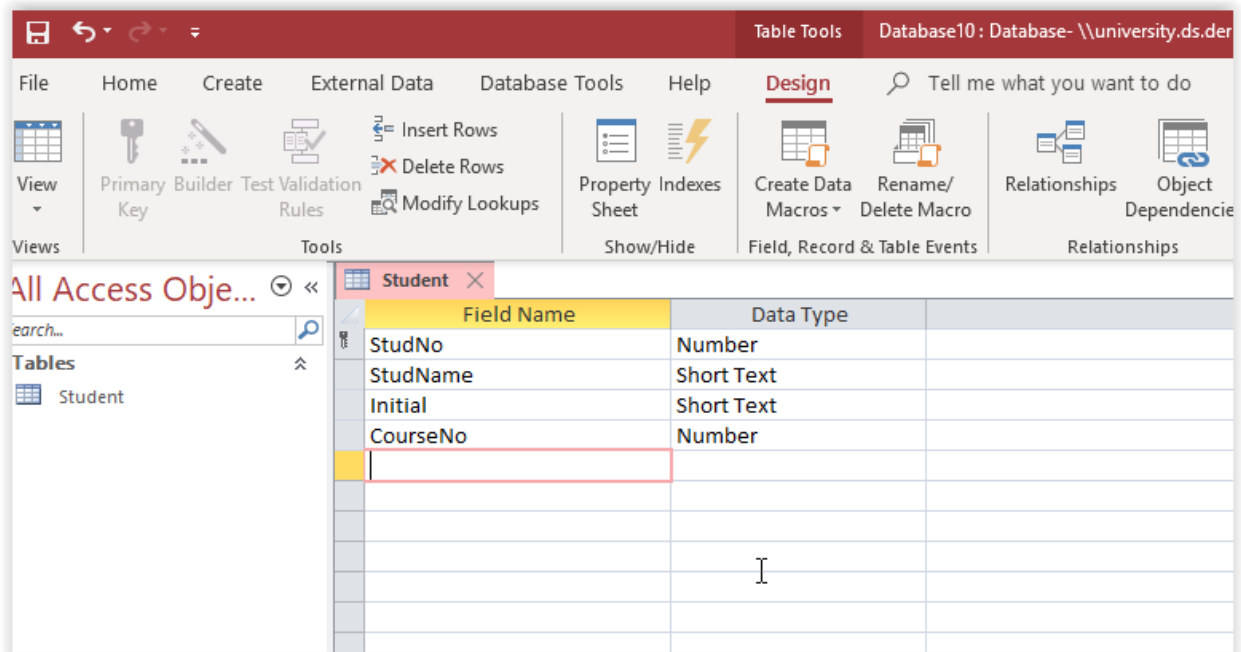
Under “Field Name”, you put the names you want for your table’s columns.

Under “Data Type”, you identify the column *domain* or type.

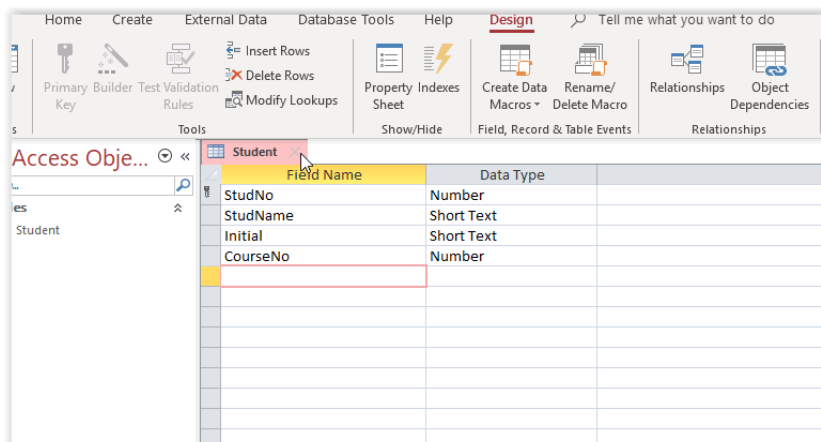
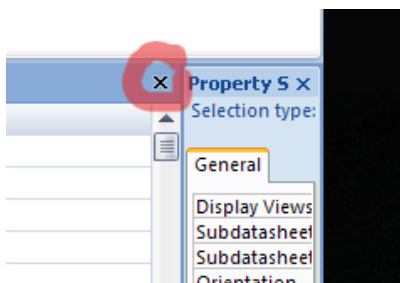
Note the “Field Properties” section toward the bottom of the window. Here, you can define additional properties for each column.

Note the “primary key” icon (a wee key) on the first column definition. By default, the first column will be the primary key. To define a new primary key, highlight the rows that belong to the primary key by holding the shift key, and clicking on the rectangle to the left of the column definition. Then click on the “Primary Key” icon in the icon bar.

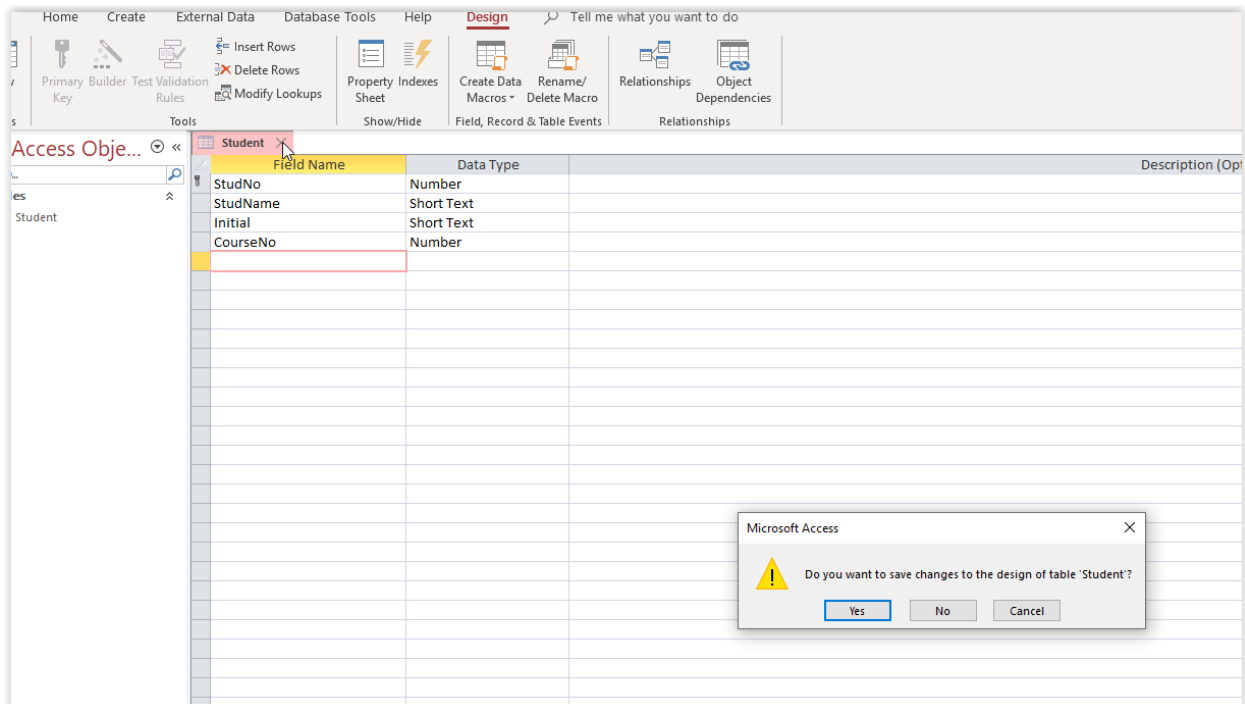
Now try defining the “Students” table as follows:



Once your table definition is complete, click on the close icon (either the ‘x’ in the upper right-hand corner of the definition area or next to the table name, depending on which version you are using) to close the table definition.

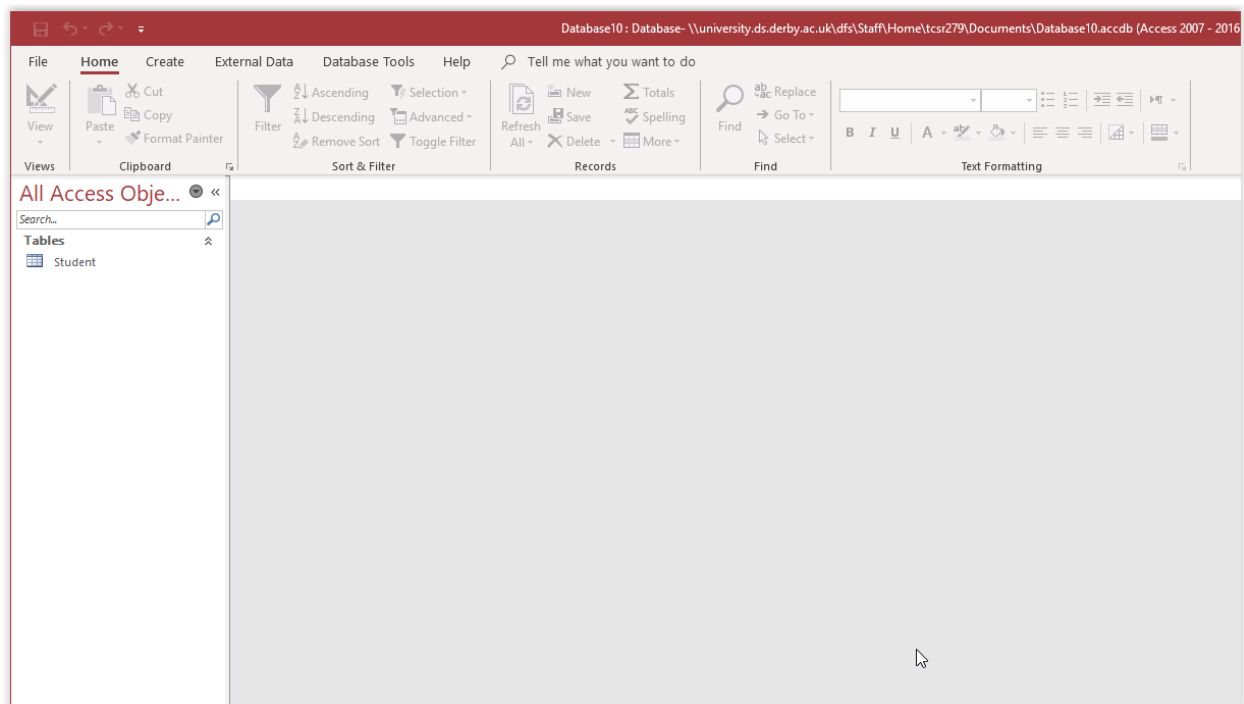


You may be asked if you wish to save your changes (i.e., your new table definition), as shown below:



Answer “Yes”.

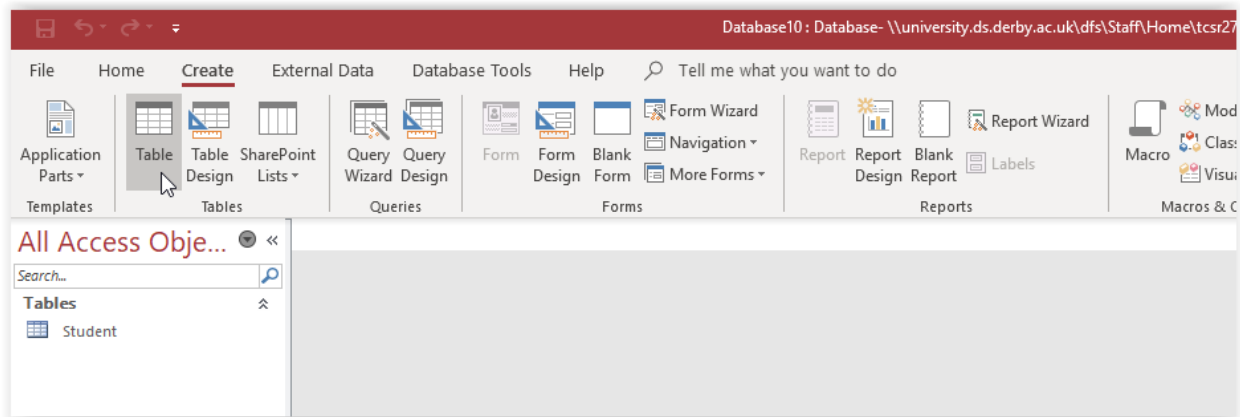
Now you should see the following:



If you wish to modify a table, right click on its name and select “Design View” from the pop-up menu. Then edit and close it as described above.

Now we'll create another table to hold "Courses" information.

To create a new table, click on the "Create" tab above the icon menu, and select "Table" from the icon menu.

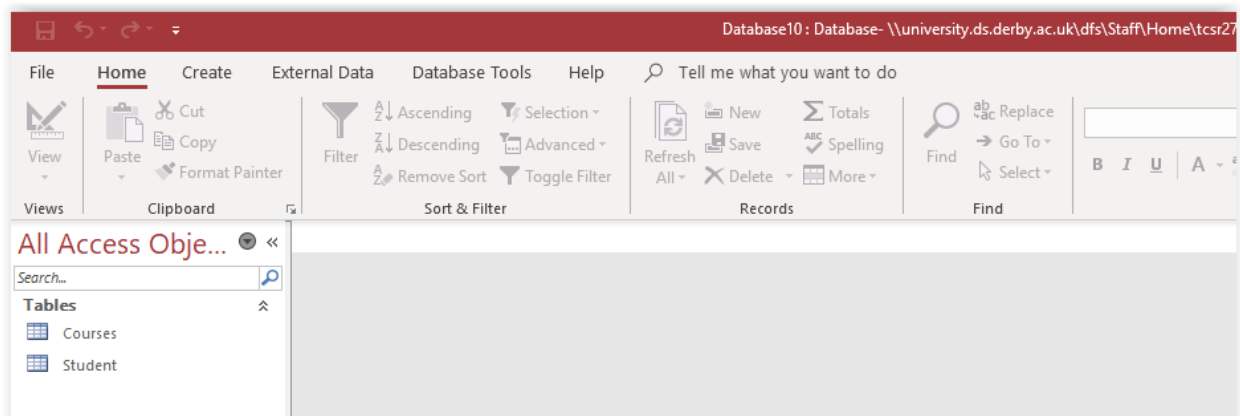


Remember to select the "View" icon and then select "Design View". When Access asks you for a table name, enter "Courses".

Now create the Courses table with the following columns:

CourseNo (primary key)	Number
Description	Short Text
Fee	Number

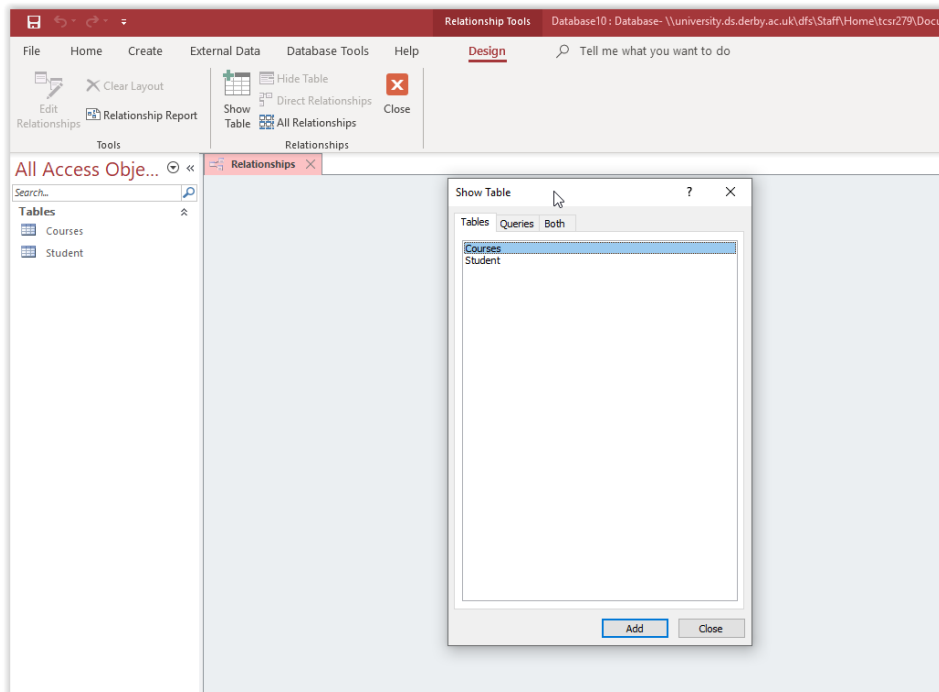
Once you've closed the "Design View" for your Courses table, you should see the following:



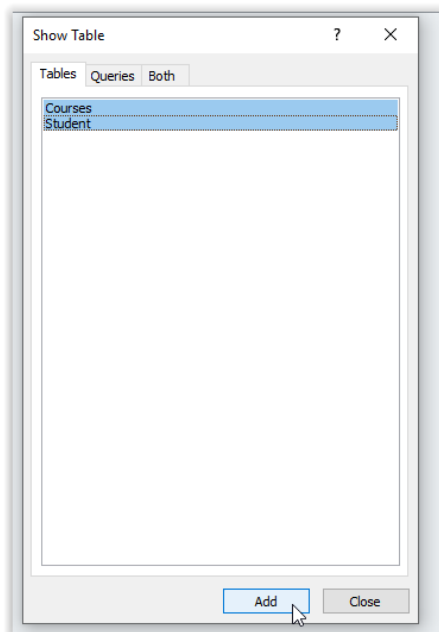
Now we need to create the appropriate foreign key constraints. Foreign key constraints cause the DBMS to enforce certain relationships between tables. For example, when entering new students into the Students table, we can use a foreign key constraint between the Students and Courses tables to ensure that only correct course numbers are used. The DBMS does this by

automatically looking up course numbers in the Courses table whenever we enter or update a row in the Students table.

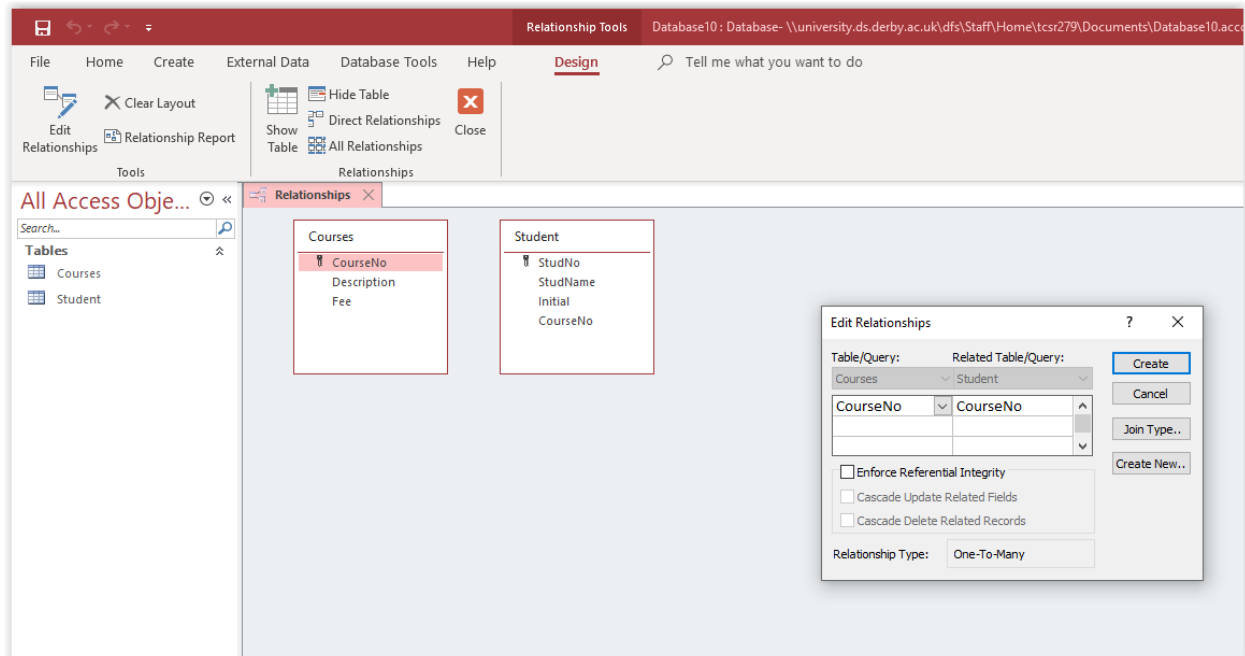
To define foreign key constraints (which Access calls Relationships), select the “Database Tools” tab, then click on the “Relationships” icon. You should see the following:



Highlight all the tables in the “Show Table” window. Then press the “Add” button. Your tables will appear as rectangles in the Relationships tab.



Now click and drag the CourseNo attribute from the Students table to the CourseNo attribute in the Courses table. When you let go of the mouse button, the following should appear:

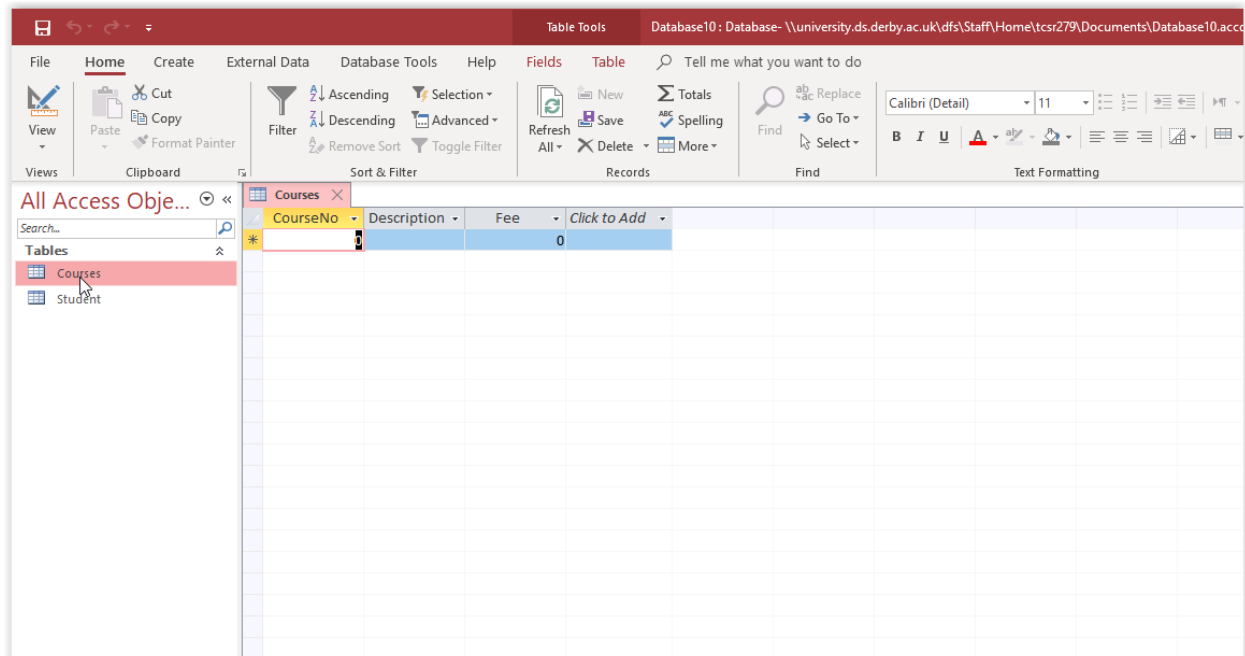


Check the three checkboxes and then click the Create button.

- Selecting “Enforce Referential Integrity” ensures that a CourseNo entered in the Students table must have a matching CourseNo in the Courses table.
- Selecting “Cascade Update Related Fields” ensures that any changes made to a CourseNo in the Courses table will automatically change any related CourseNos in the Students table.
- If you select “Cascade Delete Related Records” and you delete a Courses row, if there are any matching rows in the Students table, they will be deleted.

Now you can (finally!) enter data into your tables.

To begin data entry, double click on the table name under “All Tables” at the left-hand side of the window. We’ll start with the Courses table. You should see the following:



Enter the following:

CourseNo	Description	Fee
1423	Computing	250
1544	Information Technology	500
1792	Networks	750

Now open the Students table for data entry, and enter the following:

StudNo	StudName	Initial	CourseNo
11522	Voorhis	D	1423
11980	Jones	P	1792
12398	Smith	E	1544
14938	Smith	R	1423
16483	Salt	S	1792

Exercises

1. Remove the “Fee” column from the Courses table.
2. Modify the relationship between Students and Courses so that deleting a course does not automatically delete all the students associated with it.
3. Create a Modules table that will allow you to record the modules on a given course:

Field Name	Data Type
ModuleName	Short Text
CourseNo	Number

4. Define the primary key of the Modules table so that any given course may only list a module once, but a given module name may appear on any number of courses.
5. Create a foreign key constraint (i.e., relationship) between the Modules table and the Courses table so that deleting a course will automatically delete the modules associated with it.
6. Create an Enrolment table that will record which modules a student is studying, and the mark a student has received on the module when he/she completes it:

Field Name	Data Type
ModuleName	Short Text
CourseNo	Number
StudentNo	Number
Mark	Number

7. Define the primary key of the Enrolment table so that there may be no duplicates in the combination of ModuleName, CourseNo, and StudentNo.
8. Create a foreign key constraint between the Students table and the Enrolment table so that deleting a student will automatically delete his/her enrolment records.
9. Create a foreign key constraint so that a student can enrol on any module he/she likes (i.e., modules can be selected from any course – the student’s CourseNo is ignored), but the module must belong to the selected course. In other words, if the Networks course doesn’t have an “Intro to Databases” module, you cannot put a ModuleName of “Intro to Databases” with a CourseNo for “Networks” in the Enrolment table. (Hint: Relationships can be established between multiple columns in both tables.)

NOTE: The above is not necessarily an ideal, or even correct, model for a “real world” student information system!